

EE 522: Matrix Theory for Engineers

Course Code: EE 522

Course Name : Matrix Theory for Engineers

L-T-P-C : 3-0-0-3

Prerequisites: IC 111 Linear Algebra, or a similar course or permission from the Instructor.

Intended for: M.Tech./MS/PhD, The 3rd/4th year BTech.

Distribution: Core for M.Tech.in Communication and Signal Processing (CSP),

Elective for other postgraduates and T1hird and Final Year BTech.

Approval: 13th Senate

Course Contents

- **Background and review:** Quick recapitulation of linear system of equations, and their solutions Matrices, Detenninant, Rank, Linear Vector spaces, B is, Dimensions, Subspaces, Inner product and orthogonality. Range space and null space, Eigenvalues and eigenvectors. [4 Lectures]
- **Norms for vectors and matrices:** Vector norms and their properties, Matrix norms, Error analysis in linear systems. [4 Lectures]
- **Canonical forms, Symmetric and Hermitian matrices:** Jordon Canonical fonn, Definition, properties, and characterization of Hermitian matrices, Congruence and simultaneous diagonalization of Hermitian and symmetric matrices. [5 Lectures]
- **Perturbation theory and Eigenvalue problems:** The condition of Eigenvalues, Condition numbers and Lheir application, location and perturbation of Eigenvectors, Generalized Eigenvalue problems, Rayleigh Quotient. [5 Lectures]
- **Matrix factorization and Least square problems:** Singular value decomposition, generalized pseudoin-verscs.QR factorization, PCA, Least square problems. [4 Lectures]
- **Sparse matrices, their annlysis and algorithms:** Graphs and matrices, Linear sol vers and their complexity, Sparse Gaussian elimination, Krylov-subspace iterations, Preconditioned methods: Incomplete factorization, Sparse approximate inverses, Sparse eigenvalue and singular value problems. [5 Lectures]
- **Different types and matrices, their properties and analysis:** Irreducible, primitive. stochastic and doubly stochastic matrices; Properties of posit ive definite matrices; Sparse matrices and their analysis, Toeplitz and Circulant matrices. [5 Lectures]
- **Random matrices and their applications:** Introduction to randomness: concentration of measure, Lemma of Johnson and Lindenstrauss. Random matrices: Matrix norms, Golden-Thompson ineq uality, Non commutative Bernstein inequal ity, Lieb's theorem, Applications: matrix multiplication and m trix completion. [5 Lectures]
- **Numerical analysis and iterative methods:** Overview of iterative methods, Arnoldi iteratil)OS, Generalized minimal residual method, Lanczos iterations, Conjugate gradients, Biorthogonalization method. [5 Lectures]

Textbooks:

1. Roger A. Horn and Charles R. Johnson, **Matrix Analysis**, Cambridge university press, 2012.
2. Gene H. Golub and Charles F. Van Loan, **Matrix computations**, 3rd Edition, John Hopkins University Press, 2012.

Additional Referencetos:

1. David Lcwis, **Matrix Theory**, 3rd edition, Allied Publishers, 2014
2. T. A. Davis, **Direct Method for Sparse Linear Systems**, SIAM, 2006
3. Joel Tropp, **An Introduction to Matrix Concentration Inequalities**, 2015
4. Terence Tao, **Topics in Random Matrix Themy**, AMS, 2012
5. Lloyd N. Trefethen and David Bau Ill, **Numerical linear algebra**, Siam, 1997.
6. Alan J. Laub, **Matrix analysisfor scientists and engineers**, Siam2005.
7. Harry Dym, **Linear algebra in action**, American Mathematical Society, 2013.
8. Gilbert Strang, **Linear Algebra and its application**, 3rd Edition, Harcourth Brace Jovanovich Pubs.